

RPM Meter for Automobiles

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An efficient way to measure the RPM of a vehicle is through its ignition pulses. RPM of a vehicle is directly proportional to the rate of its ignition pulses. An ignition coil (also called spark coil) is an induc-



tion coil in an automobile's ignition system that transforms the battery's low voltage to the thousands of volts needed to create an electric spark in the spark plugs for igniting the fuel. Here we measure the rate of ignition pulses with a sensor system attached to the spark coil and thus indicate the RPM using a bar graph comprising 20 LEDs.

Circuit and working

Fig. 1 shows the circuit for RPM meter. The circuit is built around frequen-

Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	9V
TP3	Pulses (frequency proportional to RPM)
TP4	Voltage proportional to frequency

PARTS LIST

Semiconductors:

- | | |
|-------------|--|
| IC1 | - LM2917N-8 frequency-to-voltage converter |
| IC2, IC3 | - LM3914 bar/dot frequency display |
| T1 | - BD139 npn transistor |
| LED1-LED10 | - 5mm green LED |
| LED11-LED15 | - 5mm blue LED |
| LED16-LED20 | - 5mm red LED |
| D1 | - 1N4007 rectifier diode |
| ZD1 | - 9.1V zener diode |

Resisto

- | | |
|-------------|----------------------|
| R1, R7 | - 100-kilo-ohm |
| R2, R8, R10 | - 22-kilo-ohm |
| R3 | - 47-kilo-ohm |
| R4 | - 560-ohm |
| R5 | - 1-mega-ohm |
| R6 | - 470-ohm |
| R9 | - 1-kilo-ohm |
| VR1 | - 47-kilo-ohm preset |
| VR2 | - 1-kilo-ohm preset |

Capacitors:

- | | |
|----|---------------------------------|
| C1 | - 220nF ceramic disk |
| C2 | - 470nF ceramic disk |
| C3 | - 10nF ceramic disk |
| C4 | - 1 μ F, 25V electrolytic |
| C5 | - 10 μ F, 25V electrolytic |
| C6 | - 2.2 μ F, 25V electrolytic |

Miscellaneous:

- CON1 - 2-pin connector
BATT.1 - 12V, 7AH chargeable battery
- Ignition coil

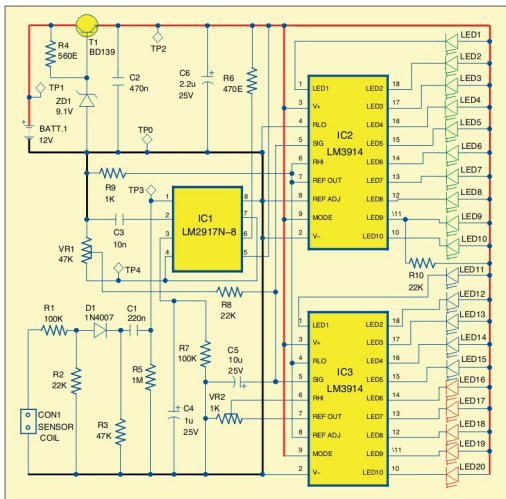


Fig. 1: RPM meter circuit

cy-to-voltage converter LM2917N-8 (IC1), bar display drivers LM3914 (IC2 and IC3), 20 LEDs (LED1 through LED20) and some other components.

The circuit has three stages—pulse detection, frequency-to-voltage conversion and display of voltage by the LED bar graph. To sense the pulses from the ignition coil, 18SWG PVC-covered household wire is wrapped around LT side of the ignition coil. The number of turns should be around 50. Induction pulses from the sensor coil must be reduced, rectified and filtered. Resistor R1 reduces their voltage level, diode D1 rectifies them and capacitor C1 filters them.

These filtered pulses are fed at pin 1 of frequency-to-voltage converter IC1. Voltage output corresponding to the frequency of pulses is available at pin 4 of IC1. The level of